Lecture 30 – Horizons II: Relativity and Spacetime Symmetries

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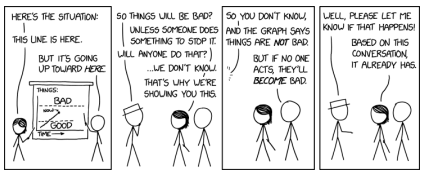
Contents

1 Relativity **2**
1.1 Symmetries 2
1.2 Noether's theorem 2

2 Symmetries in Classical Mechanics **2**

3 Symmetries in EM **3**
3.1 Problems with Maxwell's Equations with Galilean Motion 3
3.2 Lorentz Laws 4
3.3 Einstein's Theories 4



1 Relativity

The formative study of how physical laws do (and do not) depend on perspective or reference frame.

1.1 Symmetries

Sets of transformations (on Hemn space coordinates) that leave laws invariant/unchanged.

1.2 Noether's theorem

Proposed by Emmy Noether circa 1918. For every symmetry that a theory possesses, there is a corresponding conservation law.

2 Symmetries in Classical Mechanics

1. Space translation, i.e. $x' = x + \Delta x$. This is a representation of the conservation of momentum.

2. Time translation, i.e. $t' = t + \Delta t$. Representation of the conservation of time.

3. Rotation, i.e. $x' = x \cos\theta$, etc. Representation of the conservation of angular momentum.

4. Galilean transformation, meaning between inertial frames. i.e. $x' = x - vt$. Conservation of center of mass. This represents a simplification of the

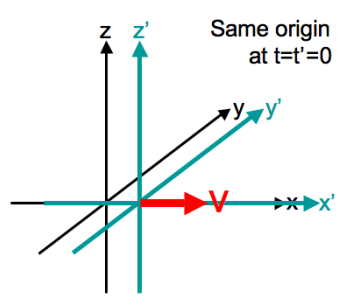


Figure 1: Galilean transformation

change in inertial reference frames. From the above example:

$$x' = x - vt$$

$$y' = y$$

$$z' = z$$

$$t' = t$$

Situations involving accelerations are much the same.

$$a'_x = v_x - a_x t$$

$$a'_y = a_y$$

$$a'_z = a_z$$

$$t' = t$$

Additionaly, because $\vec{a}' = \vec{a}$, then $\vec{F} = m\vec{a}$ may be used in these translations as well.

3 Symmetries in EM

- Situation is more complicated as $\vec{E}, \vec{B}$ mix under Galilean transformation. Consider a charge q in two frames:
  1. In a frame where q is stationary, we find only $\vec{E}$.
  2. In a frame where q is moving relative to the frame itself, we find both $\vec{E}$ and $\vec{B}$.

3.1 Problems with Maxwell's Equations with Galilean Motion

Unlike Newton's laws, Maxwell's laws are not invariant under Galilean transformations. This seems to imply a greater reference frame of the universe at large. This theoretical frame was named the Aether.

But wait, doesn't that violate the whole concept of universal relativity? It does. This is why the idea of special relativity was developed; transformations

that allowed for the transformation of EM fields were developed by Hendrik Lorentz.

3.2 Lorentz Laws

$$x' = \gamma(x - ut)$$

$$y' = y$$

$$z' = z$$

$$t = \gamma(t - \frac{ux}{c^2})$$

Where,

$$\gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}$$

The solution to the issue of reconciling Maxwell's laws with Galilean transformations wasn't through changing Maxwell's equations – rather, it was through changing our definitions of spacetime.

3.3 Einstein's Theories

1. There is no aether – all frames are equivalent. Issues can be likely reconciled through the Lorentz transformations.

2. $E = mc^2$